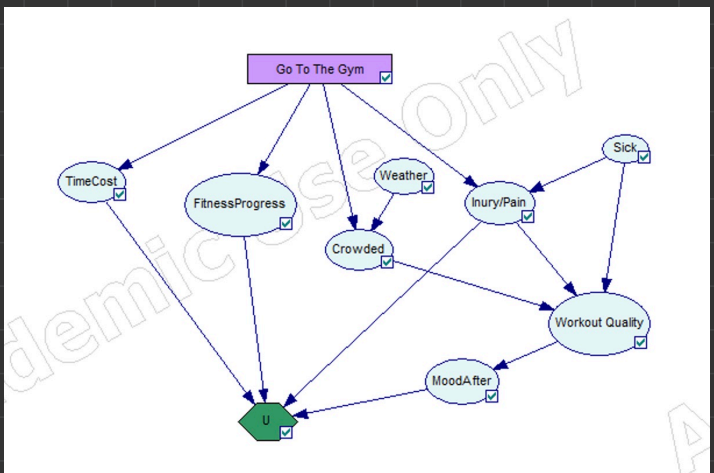
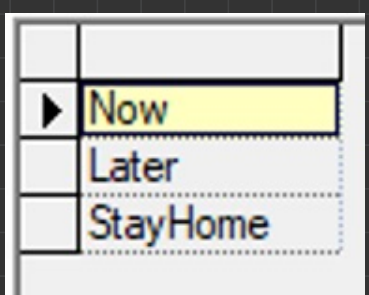
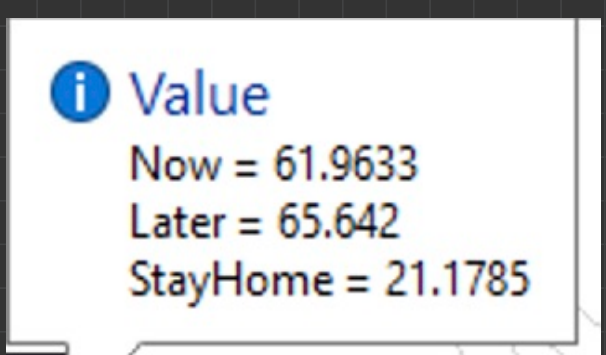
# Assignment 5




Screenshot showing the model




The different choices for going to the gym




output example showing values for going to gym now, go later or stay at home

Table from FitnessProgress showing the progress of fitness (poor, same, good) based on the choices go to gym now, go later or stay at home

| Go To The Gym | Now | Later | StayHome |
|---|---|---|---|
| Poor | 0.01 | 0.01 | 0.95 |
| Same | 0.14 | 0.09 | 0.05 |
| Better | 0.85 | 0.9 | 0 |

Table from MoodAfter showing how the mood (worse, same, better) are after a workout based on the quality of the workout

| Workout Quality | Poor | OK | Great |
|---|---|---|---|
| Worse | 0.5 | 0.1 | 0.01 |
| Same | 0.35 | 0.3 | 0.04 |
| Better | 0.15 | 0.6 | 0.95 |

Example from utility node showing fitnessProgress better, MoodAfter better, timecost and injury/pain

| FitnessProgress | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| MoodAfter | Better | | | | | | | |
| TimeCost | High | | Low | | Medium | | High | |
| Inury/Pain | No | Yes | No | Yes | No | Yes | No | Yes |
| Value | 35 | -85 | 100 | -20 | 90 | -30 | 75 | -45 |

## Decision node

**Go To The Gym**: represents the choice between going now, going later, or staying home.

## Chance nodes (uncertain variables)

The following uncertain variables are included in the model:
**Weather**: affects gym crowding conditions.

**Sick**: represents whether I am sick or healthy.

**Crowded**: represents how crowded the gym is.

**Injury/Pain**: represents the probability of injury or pain.

**Workout Quality**: represents the quality of the workout session.

**MoodAfter**: represents how I feel after the decision.

**TimeCost**: represents the time required for the activity.

**FitnessProgress**: represents the fitness benefit gained.

These variables are treated as uncertain at the time of decision and are observed after the decision is made.
The relationships between variables are represented using directed edges. For example:
The decision influences time cost, fitness progress, and injury risk.
Workout quality influence mood and overall utility.
Weather influences crowding, which affects workout quality.

## Utility function

The success of each decision is measured using a utility node, which represents overall desirability. The utility depends on:

FitnessProgress (positive effect)
MoodAfter (positive or negative effect)
TimeCost (negative effect)
Injury/Pain (negative effect)

## Quantification of uncertainty

All uncertain variables were assigned probability distributions based on personal experience and reasonable assumptions. For example:

- Going to the gym increases the probability of higher fitness progress.
- Staying home results in low fitness progress but lower injury risk and time cost.
-

These probability distributions allow the model to calculate expected utility for each decision alternative.

## Decision analysis and results

After defining the model and utility values, the expected utility of each decision was calculated using GeNIe.
The model recommends the decision alternative with the highest expected utility. In general, going to the gym results in higher expected utility due to fitness progress and improved mood. However, under certain conditions, such as being sick or having a high risk of injury, staying home may result in higher utility.
This demonstrates that the decision is non-trivial and depends on uncertain factors.